

Education

University of Illinois Urbana-Champaign (UIUC) 2026.08 – 2028.06

M.Eng. in Electrical and Computer Engineering, Grainger College of Engineering

China Jiliang University 2022.09 – 2026.06

B.Eng. in Optoelectronic Information Science and Engineering, Liangxin College (**Honors College**)

GPA 93/100, Rank 1/223; Core Courses: Mathematical Analysis (96), Linear Algebra (95), Probability Theory (98)

Internship

ByteDance | Data-aml | Automotive LLM/VLM Algorithm Intern 2026.01 – Present

Project 1: Built an image captioning system, supporting autonomous driving semantic understanding and cockpit human-machine interaction.

- **SFT Cold Start:** Vectorized images with an embedding model and applied **K-Means** to select cluster-center samples; converted GT JSON annotations into natural-language captions with a seed model, enabling an **SFT** cold start from limited high-quality data.
- **RLVR + GRPO:** Designed a two-stage **RLVR + GRPO** reward pipeline: first used a decoupled LLM to parse generated captions back into JSON and perform field-level TP/FN/FP matching against GT; then generated MCQs from GT/images and used **decoupled** LLM-based VQA accuracy as the reward signal.

Project 2: Fine-tuned doubao-embedding-vision and seed-series VLMs for autonomous driving scenarios, improving road element recognition capability.

- **Pipeline:** Built a fine-tuning workflow covering data processing, training, and evaluation; established multidimensional metrics including single-label precision/recall and model-forgetting detection. Fine-tuned models with **FullSFT** and **LoRA**, and designed ablation experiments to optimize positive/negative sample ratios and **hard negative** construction.
- **Performance Improvement:** Improved the embedding model's average single-label Precision@K from 0.196 to **0.839**; achieved **59.73%** Micro-F1 and **54.67%** Macro-F1 on the VLM model.

Project 3: Built an end-cloud integrated offline LLM rejection and instruction-parsing pipeline for always-on, wake-word-free intelligent cockpit, distinguishing human-machine and human-human conversations to reduce false wakeups and alleviate cloud-side Always-On concurrency pressure.

- **End-Cloud Integration:** **SFT + ToolRL** offline LLM as a high-precision, low-recall local rejection/FunctionCall parser; the cloud side complements it with high-recall rejection and a two-stage LLM fallback, with Advisor performing final arbitration. Achieved offline rejection latency **<100ms**, instruction accuracy **95%**, and rejection accuracy **98%**.
- **Fine-Grained Labels + Dual-Mode Post-Processing:** Refined rejection labels and designed **strict/normal dual-mode** post-processing to maximize rejection in multi-passenger scenarios and recall in single-user/active sound-zone scenarios; introduced **dynamic semantic library injection** for hot instruction updates.
- **Evaluation:** Built multi-scenario test sets and multidimensional metrics to support iteration and strategy optimization.

XPeng Motors | Intelligent Cockpit Center | VLM Foundation Model Intern 2025.10–2026.01

Project 1: Optimized an image captioning model to improve information density and accuracy while reducing hallucinations.

- **Algorithm Design:** Designed a **GRPO training framework based on decoupled VQA rewards** to reduce costly annotation dependence, data diversity limits, and judge-model reward hacking. Through **RLVR**, maximized downstream VQA accuracy, guiding the model to generate denser, more accurate, and hallucination-free Captions.
- **Data Engineering:** Built a dual-verification data pipeline to filter invalid samples with information leakage, enforce vision-dependent answering, and significantly improve training data **signal-to-noise**.

Project 2: Optimized VLM's spatial capability intelligence to improve spatial relationship understanding and reasoning.

- **Benchmarking:** Selected spatial benchmarks, aligned metrics across open-source models, and established the baseline.
- **Model Training:** Collected and curated high-quality data; **SFT cold start** for spatial instruction following and **GRPO** with custom reward design for spatial-reasoning, achieving **SOTA** on CountBench and RefSpatial Bench.

Trusight Technology | Trusight AI Research | Computer Vision R&D Intern 2025.07 – 2025.09

Project: Optimized object tracking algorithms for robustness under occlusion, fade-in/out and related tracking-loss scenarios.

- **Solution Iteration:** Led the transition from Tracking-by-Detection → Tracking-by-Template → **point tracking**, adopting point trackers as the core approach and achieving **more stable cross-frame association** across video tasks.
- **Engineering Deployment:** Benchmarked, optimized, and deployed point trackers including CoTrack, TAPNext, and Locotrack in real-world scenarios, improving overall tracking accuracy by **10%+** and computational **efficiency** by **2x**.

Academic Projects

3DMM-FLUX: Instruction-Guided Facial Expression Editing | Undergraduate Thesis

[🔗Code] [🔗Dataset] [🤗Hugging Face]

2026.03 – 2026.05

Project: Proposed an instruction-driven expression editing framework to address **identity drift** and **fine-grained control** in diffusion-based editing, using **FLUX** as the generative backbone and **DECA**, **FLAME**, **Nvdiffraft** as the geometric bridge for an end-to-end differentiable pipeline: “natural language instruction + reference image → target 3DMM parameters → differentiable rendering control map → diffusion generation.”

- **Core Innovation:** Designed a **Conditional DECA Encoder** that deeply fuses text Embeddings as K/V and image features as Q, predicting only **50d expression parameters + 3d jaw rotation parameters**, while inheriting the remaining 183d identity/texture/lighting parameters from reference image to eliminate identity drift; proposed (**RCG**), a CFG-style inference strategy that amplifies “reference control vs target control” branch differences, achieving **AU Acc+3.8pp** with zero training cost.
- **Training Strategy:** Adopted a two-stage training strategy of “**parameter-space pretraining + end-to-end fine-tuning**”: first aligned 3DMM parameters with **L2 supervision**, then jointly optimized the full pipeline with **diffusion denoising loss**, balancing convergence stability and generation quality.
- **Evaluation:** Outperformed the FLUX base model on the test set, achieving **AU Acc↑22.2pp to 81.4%**, **ID Sim↑0.072 to 0.824**, and **FID↓8.7 to 15.6**.

AI Scientist Assistant | Institute of Software, Chinese Academy of Sciences

2025.09 – 2025.10

- Analyzed AI Scientist, an automated scientific paper generation system, covering idea generation, novelty retrieval, automated experiments, paper writing, and review; clarified prompt structures and stage-wise interaction patterns.
- **AIGC Text Detection:** Built a **RoBERTa + AdaLoc** sentence-level detector with **sliding-window voting** and **overlapping-window aggregation**, achieved **92%** sentence-level accuracy in distinguishing human from AI.

Illegal Advertisement Text Classification Based on NLP

2024.08 – 2024.09

- Compared the recognition efficiency of three text vectorization methods, One-Hot, TF-IDF, and Word2Vec, in long-text illegal advertisement classification; trained an SVM classifier with GridSearchCV tuning, improving accuracy from 69.8% to 88.3%. Contributed to 1 EI conference paper at NLPIR 2024 and 1 software copyright.

Competition Experience

Kaggle: Student Math Misconception Analysis | Silver Medal

2025.09

Project: A text classification task predicting answer correctness, misconception presence, and misconception type from math problems, student answers, and explanations, evaluated by **MAP@3**.

- **Feature Engineering:** Built key contextual features tailored to the data characteristics, significantly improving the model’s understanding of error attribution and increasing the test-set score by **20%**.
- **Dual-path fine-tuning:** (1) **FullSFT** of DeepSeek-Math-7B, Qwen-8B, and Gemma2-9B; (2) innovatively reformulated classification as generation and applied **LoRA** on the Qwen2.5-32B. Finally designed a **weighted ensemble strategy** to fuse four-model predictions and combine strengths across paradigms.
- **Inference:** Applied **GPTQ 4bit quantization** and deployed with **vLLM** for low-latency, high-throughput inference.

National Mathematical Modeling Competition | Provincial First Prize | Team Leader

2024.09

- Led the team to complete a crop planting strategy optimization problem, responsible for data preprocessing, building a multi-objective optimization model, and solving for the optimal planting plan.

China International College Students’ Innovation Competition & National Undergraduate Physics Experiment Competition | Provincial Silver Awards | Team Leader

2024.05 – 2025.05

- Participated in the R&D of a power-compensation battery isothermal calorimeter, responsible for system construction and data processing. **Authorized 3 software copyrights, 1 invention patent, and 1 utility model patent.**

Honors

- **Scholarships and Honors:** 2025 Zhejiang Provincial Government Scholarship, 2026 Guangzhou High-Speed Railway Metrology Scholarship, First-Class School Scholarship (4×), Merit Student (3×)
- **Language Proficiency:** IELTS 7.0 (L6.5, R9.0, W7.0, S6.0), GRE 327 (V157, Q170, W3.5).